

[[CS 726: Advanced machine learning.]]

Now: probabilistic Foundations of AI (CS 791)

Wanted to discuss a quick outline of topics covered in this course - for an idea in future.

Phase - I: classical probabilistic modelling.

1. Probabilistic Graphical Models-

* Directed graphical models (Bayesian Networks)

- Representation of joint distribution. BNs
- Conditional Independence. CI
- D-separation ; Factorization.

* Undirected graphical models. (Markov Random Fields)

what buzz words!

MRFs

- potentials, Separation
- Gibbs distribution ☉
- Ising & Potts model?
- Log-linear models
- Relation with Bayesian Networks?

* Factor Graphs. means?

* Conditional Random Fields (CRFs)

- feature based potentials-
- structured predictions. ? means?

2. Inference in graphical Model

graph models...
what's the
big deal?

- Variable elimination.
- message passing; Sum-product
- Junction trees. ???

3. Learning graphical models

- parameter learning
- Learning with Latent variables?
- Learning CRFs ...

4) Sampling and approximate inference...

- Forward Sampling
- Importance Sampling
- MCMC !! markov chain monte carlo
 - └ Gibbs sampling
 - └ Langevin Dynamics. hum..
- Variational Inference...

Phase-II: Deep probabilistic Generative Models.

* Deep Latent Representation Models.

- Variational Autoencoders (VAEs)
 - Generative Adversarial Networks (GANs)
 - Normalizing Flows
 - Diffusion Models. (Some paper also shared here)..
? for A-2 ig.
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Phase-III: Modern Foundation Models.

* Transformers

- Transformer models
- Representation Analysis.
- State space models (introduction)

* LLM as probabilistic models !! pretty latest?

- LLM as bayesian Network
- Greedy decoding
- Sampling
- Grammar-Constrained decoding
- Speculative decoding.

* Additional topics

- state space models.
 - parameter-efficient Fine-tuning (PEFT)
 - in-context learning
hmm... ?
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Phase-IV : Beyond classical prediction

* Gaussian Processes ! interesting topic.

* Bayesian Optimization (tutorial)

* Time series Models → Always felt whether this was relevant for trading or not....

- Deep AR
- Gaussian Copula processes

* Causal Inference → this also, a big topic I think..?

- Causal graphs
- Identifiability
- Estimation from observable data.. hmm
- Counterfactual reasoning...
- Counterfactual explanation.

x ————— x

Q. What's the big deal about graphical models?

- "How do we represent probability distribution over many interacting variables?"

- the entire reason they exist.

- 1) How to represent distribution compactly
- 2) Learn exponentially smaller.
- 3) Inference on probabilities is NP-hard on large graphs;
so approximate inference concepts!

* Deep learning has replaced hand-designed graphical models

But the underlying ideas did not disappear.

- introduce central abstractions of
 - factorization
 - conditional independence
 - probabilistic inference.

* These again occur in VAE, diffusion models, transformers, Gaussian Processes, Causality.

Q) More on sampling & approximate inference...

- Big models, exact inference is not feasible.

- Here problem is NOT learning;
its computing.

Hence, 2 ways.

1) Optimization: Approximate the distribution to simpler one.

- mean Field
- Variational Inference.

2) Sampling: don't compute, draw samples.

- MCMC
- Gibbs sampling
- Langevin Dynamics

- many latest models, etc, rely on this.

